

Muhammad Sameer Akram Mughal

Aspiring **AI Engineer** | **Machine Learning Engineer**

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Artificial Intelligence graduate with hands-on experience building **machine learning and deep learning solutions** through academic projects. Led a Final Year Project on speech-based depression screening using deep learning (FYP-I), later extended to a multi-modal approach using text, audio, and facial features (FYP-II – TriDep), and developed MindBridge-RAG, a safety-aware RAG chatbot for student exam stress support. Skilled in **Python, TensorFlow, PyTorch, Scikit-learn, OpenCV, and SQL**.

AREAS OF EXPERTISE

Python • SQL • C++ • TensorFlow • PyTorch • Keras • Scikit-learn • Deep Learning • Computer Vision • OpenCV • Pandas • NumPy • RAG • Git • GitHub • Google Colab • Jupyter Notebook • Problem-solving • Teamwork • Time Management • Continuous Learning

PROJECTS

Sequential Deep Learning Models for Emotion-Based Depression Detection | Final Year Project I, UMT Sialkot

- Built a two-stage speech emotion recognition (SER) and depression-inference pipeline using deep learning models (LSTM, Bi-LSTM, 1D-CNN, and a CNN-LSTM hybrid) trained on the CREMA-D speech emotion database.
- Engineered acoustic features (13 MFCCs, ZCR, RMS energy) with audio preprocessing and augmentation (noise addition, time-stretching, time-shifting) to improve model robustness.
- Delivered strong performance on binary depression-tendency inference with the best-performing C-LSTM model, using TensorFlow/Keras, Librosa, and Scikit-learn.

TriDep – Multi-Modal Depression Screening | Final Year Project II, UMT Sialkot

- Built a multimodal depression screening system integrating text, audio, and facial features using deep learning models.
- Combined multiple modalities into a unified pipeline to improve robustness of mental-health risk detection.

MindBridge-RAG | Academic Project

- Developed a safety-aware Retrieval-Augmented Generation (RAG) chatbot to support students facing exam stress.
- Designed safety guardrails to ensure appropriate, context-aware responses for sensitive student wellbeing queries.

Denosing Autoencoder | Academic Project

- Implemented a **deep learning** (TensorFlow/Keras) autoencoder to remove noise from MNIST and Fashion-MNIST image datasets.
- Evaluated reconstruction quality to benchmark model performance across varying noise levels.

Course Registration Database | Academic Project

- Designed a normalized SQL database schema for course registration and student management.
- Structured tables to support efficient querying, data integrity, and scalability.

EDUCATION

University of Management and Technology (UMT), Sialkot

Completed July 2026

- BS Artificial Intelligence — CGPA: 3.50/4.00 (after 7th semester); final result awaited.

Govt. Murray College, Sialkot — Intermediate (A+)

Govt. Pilot Higher Secondary School, Sialkot — Matriculation (A+)

COURSES & CERTIFICATIONS

- CS50P: CS50's Introduction to Programming with Python – Harvard University (2026)
- Google Data Analytics Professional Certificate (In Progress)
- The AI Filmmaking Pipeline Certificate – Higgsfield Academy (July 2026)

ACHIEVEMENTS

- Dean's Merit Award
- Gold Medalist – Govt. Murray College
- PEEF Scholarship
- Rehmat-ul-Alameen Scholarship

LANGUAGES

Urdu (Native) • Punjabi (Native) • English (Professional Working Proficiency)